

Problem 1. Area of a Parametrized Region. Given a region D in $\mathbb{R}^2$, the area is

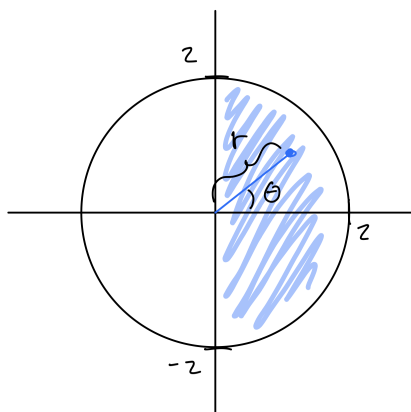
$$\text{Area}(D) = \iint_D 1 \, dx dy.$$

For each of the following problems you should (1) draw the region, (2) find a parametrization, (3) use your parametrization to compute the area.

- (a) The half-circle satisfying $x^2 + y^2 \leq 4$ and $x \geq 0$. [Hint: Use polar coordinates.]
- (b) The region satisfying $x^2 + y^2 \leq 4$ and $x \geq 1$. [Hint: Don't use polar coordinates. You will need the antiderivative

$$\int 2\sqrt{4-x^2} \, dx = x\sqrt{4-x^2} + 4\arcsin(x/2).]$$

(a): In this case D is the right half of a circle of radius 2:

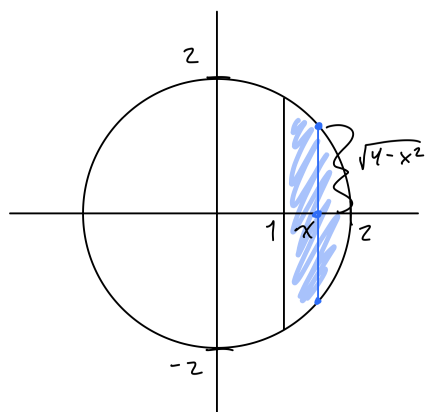


We parametrize D using polar coordinates with $0 \leq r \leq 2$ and $-\pi/2 \leq \theta \leq \pi/2$. The area is

$$\begin{aligned} \iint 1 \, dx dy &= \iint r \, dr d\theta \\ &= \int_{-\pi/2}^{\pi/2} 1 \, d\theta \int_0^2 r \, dr \\ &= [\theta]_{\theta=-\pi/2}^{\theta=\pi/2} \left[\frac{1}{2} r^2 \right]_{r=0}^{r=2} \\ &= \left[\frac{\pi}{2} + \frac{\pi}{2} \right] \left[\frac{1}{2} 2^2 \right] \\ &= 2\pi. \end{aligned}$$

Note that this agrees with the formula $\pi(2)^2 = 4\pi$ for the area of the full circle.

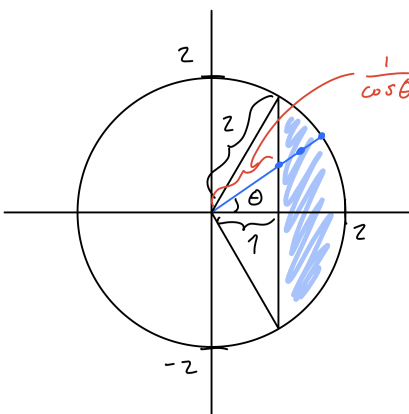
(b): Here is a picture of the region D :



We can parametrize this region by $1 \leq x \leq 2$ and $-\sqrt{4-x^2} \leq y \leq \sqrt{4-x^2}$. The area is

$$\begin{aligned}
 \iint 1 \, dx \, dy &= \int_1^2 \left(\int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} 1 \, dy \right) dx \\
 &= \int_1^2 [y]_{y=-\sqrt{4-x^2}}^{y=\sqrt{4-x^2}} dx \\
 &= \int_1^2 2\sqrt{4-x^2} \, dx \\
 &= \left[x\sqrt{4-x^2} + 4 \arcsin(x/2) \right]_{x=1}^{x=2} \\
 &= 2\sqrt{4-2^2} + 4 \arcsin(2/2) - 1\sqrt{4-1^2} - 4 \arcsin(1/2) \\
 &= 0 + 4 \arcsin(1) - \sqrt{3} - 4 \arcsin(1/2) \\
 &= 0 + 4(\pi/2) - \sqrt{3} - 4(\pi/6) \\
 &= 4\pi/3 - \sqrt{3}.
 \end{aligned}$$

Remark: It is also possible (but more difficult) to parametrize this region using polar coordinates. Consider the following picture:



The picture shows that $-\pi/3 \leq \theta \leq \pi/3$ and $1/\cos \theta \leq r \leq 2$, so the area is

$$\iint 1 \, dx \, dy = \iint r \, dr \, d\theta$$

$$\begin{aligned}
&= \int_{-\pi/3}^{\pi/3} \left(\int_{1/\cos\theta}^2 r \, dr \right) d\theta \\
&= \int_{-\pi/3}^{\pi/3} \left[\frac{1}{2} r^2 \right]_{r=1/\cos\theta}^{r=2} d\theta \\
&= \int_{-\pi/3}^{\pi/3} \left[2 - \frac{1}{2 \cos^2 \theta} \right] d\theta \\
&= \left[2\theta - \frac{\tan \theta}{2} \right]_{\theta=-\pi/3}^{\theta=\pi/3} \\
&= 2 \left[\frac{\pi}{3} + \frac{\pi}{3} \right] - \frac{1}{2} [\tan(\pi/3) - \tan(-\pi/3)] \\
&= 2 \left[\frac{2\pi}{3} \right] - \frac{1}{2} [\sqrt{3} + \sqrt{3}] \\
&= 4\pi/3 - \sqrt{3}.
\end{aligned}$$

The fact that we got the same answer each time means that the calculations are probably correct. This problem can also be solved without Calculus:

https://en.wikipedia.org/wiki/Circular_segment#Arc_length_and_area

Problem 2. Center of Mass of a 2D Region. Let D be the region parametrized by $0 \leq x \leq 2$ and $x \leq y \leq 5x - 2x^2$. Think of D as a solid with mass density 1.

- Compute the total mass $M = \iint_D 1 \, dx dy$.
- Compute the moments $M_x = \iint_D x \, dx dy$ and $M_y = \iint_D y \, dx dy$.
- Compute the center of mass.
- Draw the region and its center of mass.

(a): The mass (i.e., the area) of the region D is

$$\begin{aligned}
M &= \iint_D 1 \, dx dy \\
&= \int_0^2 \left(\int_x^{5x-2x^2} 1 \, dy \right) dx \\
&= \int_0^2 (5x - 2x^2 - x) \, dx \\
&= \int_0^2 (4x - 2x^2) \, dx \\
&= \left[4 \frac{1}{2} x^2 - 2 \frac{1}{3} x^3 \right]_0^2 \\
&= 4 \frac{1}{2} 2^2 - 2 \frac{1}{3} 2^3 \\
&= 8/3.
\end{aligned}$$

(b): The moment in the x direction is

$$M_x = \iint_D x \, dx dy$$

$$\begin{aligned}
&= \int_0^2 \left(\int_x^{5x-2x^2} x \, dy \right) dx \\
&= \int_0^2 [xy]_{y=x}^{y=5x-2x^2} dx \\
&= \int_0^2 [x(5x-2x^2) - x^2] dx \\
&= \int_0^2 [4x^2 - 2x^3] dx \\
&= \left[4\frac{1}{3}x^3 - 2\frac{1}{4}x^4 \right]_0^2 \\
&= 4\frac{1}{3}2^3 - 2\frac{1}{4}2^4 \\
&= 8/3.
\end{aligned}$$

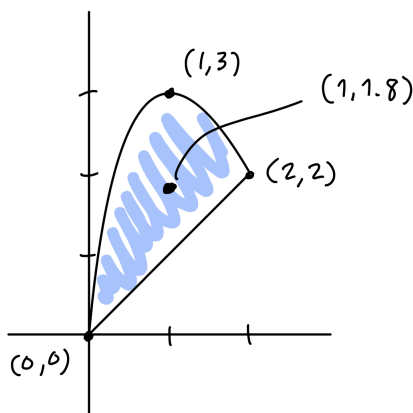
It is just a coincidence that $M_x = M$. The moment in the y direction is

$$\begin{aligned}
M_y &= \iint_D y \, dx \, dy \\
&= \int_0^2 \left(\int_x^{5x-2x^2} y \, dy \right) dx \\
&= \int_0^2 \left[\frac{1}{2}y^2 \right]_{y=x}^{y=5x-2x^2} dx \\
&= \frac{1}{2} \int_0^2 [(5x-2x^2)^2 - x^2] dx \\
&= \frac{1}{2} \int_0^2 [4x^4 - 20x^3 + 24x^2] dx \\
&= \frac{1}{2} \left[4\frac{1}{5}x^5 - 20\frac{1}{4}x^4 + 24\frac{1}{3}x^3 \right]_0^2 \\
&= \frac{1}{2} \left(4\frac{1}{5}2^5 - 20\frac{1}{4}2^4 + 24\frac{1}{3}2^3 \right) \\
&= 24/5.
\end{aligned}$$

(c): The center of mass is

$$(\bar{x}, \bar{y}) = \left(\frac{M_x}{M}, \frac{M_y}{M} \right) = \left(\frac{8/3}{8/3}, \frac{24/5}{8/3} \right) = \left(1, \frac{9}{5} \right) = (1, 1.8).$$

(d): Here is a picture:



Problem 3. Polar Coordinates. Let $x = r \cos \theta$ and $y = r \sin \theta$. We already know that

$$\frac{\partial(x, y)}{\partial(r, \theta)} = \det \begin{pmatrix} x_r & x_\theta \\ y_r & y_\theta \end{pmatrix} = r.$$

The general theory predicts that we must also have

$$\frac{\partial(r, \theta)}{\partial(x, y)} = \det \begin{pmatrix} r_x & r_y \\ \theta_x & \theta_y \end{pmatrix} = \frac{1}{r}.$$

Check that this is true. [Hint: $r = \sqrt{x^2 + y^2}$ and $\theta = \arctan(y/x)$.]

First we compute the derivatives using the one variable chain rule:

$$r_x = (1/2)(x^2 + y^2)^{-1/2}(2x) = x(x^2 + y^2)^{-1/2},$$

$$r_y = (1/2)(x^2 + y^2)^{-1/2}(2y) = y(x^2 + y^2)^{-1/2},$$

$$\theta_x = 1/(1 + (y/x)^2)(-y/x^2),$$

$$\theta_y = 1/(1 + (y/x)^2)(1/x).$$

The formulas for θ_x and θ_y can be simplified using $1/(1 + (y/x)^2) = x^2/(x^2 + y^2)$ to get

$$\theta_x = x^2/(x^2 + y^2)(-y/x^2) = -y/(x^2 + y^2),$$

$$\theta_y = x^2/(x^2 + y^2)(1/x) = x/(x^2 + y^2).$$

Then we compute the determinant:

$$\begin{aligned} \det \begin{pmatrix} r_x & r_y \\ \theta_x & \theta_y \end{pmatrix} &= r_x \theta_y - r_y \theta_x \\ &= x(x^2 + y^2)^{-1/2} x/(x^2 + y^2) + y(x^2 + y^2)^{-1/2} y/(x^2 + y^2) \\ &= x^2(x^2 + y^2)^{-3/2} + y^2(x^2 + y^2)^{-3/2} \\ &= (x^2 + y^2)(x^2 + y^2)^{-3/2} \\ &= (x^2 + y^2)^{-1/2} \\ &= 1/\sqrt{x^2 + y^2} \\ &= 1/r. \end{aligned}$$

That was weirdly complicated, but we got the right answer.

Problem 4. Center of Mass of a 3D Region. Let D be the tetrahedron with vertices $(0, 0, 0)$, $(1, 0, 0)$, $(0, 1, 0)$ and $(0, 0, 1)$. Think of D as a solid with constant mass density 1. This region can be parametrized by $0 \leq x \leq 1$, $0 \leq y \leq 1 - x$ and $0 \leq z \leq 1 - x - y$.

- (a) Compute the total mass $M = \iiint_D 1 \, dx \, dy \, dz$.
 (b) Compute the moments

$$M_x = \iiint_D x \, dx \, dy \, dz, \quad M_y = \iiint_D y \, dx \, dy \, dz, \quad M_z = \iiint_D z \, dx \, dy \, dz.$$

[Hint: There might be a shortcut.]

- (c) Compute the center of mass.

(a): The total mass (i.e., the volume) is

$$\begin{aligned} M &= \iiint_D 1 \, dx \, dy \, dz \\ &= \int_0^1 \left(\int_0^{1-x} \left(\int_0^{1-x-y} 1 \, dz \right) dy \right) dx \\ &= \int_0^1 \left(\int_0^{1-x} (1 - x - y) dy \right) dx \\ &= \int_0^1 \left[y - xy - \frac{1}{2}y^2 \right]_0^{1-x} dx \\ &= \int_0^1 \left[(1-x) - x(1-x) - \frac{1}{2}(1-x)^2 \right] dx \\ &= \int_0^1 \left[\frac{1}{2}x^2 - x + \frac{1}{2} \right] dx \\ &= \left[\frac{1}{2} \frac{1}{3} x^3 - \frac{1}{2} x^2 + \frac{1}{2} x \right]_0^1 \\ &= \frac{1}{2} \frac{1}{3} - \frac{1}{2} + \frac{1}{2} \\ &= 1/6. \end{aligned}$$

(b): Because the shape D is symmetric under permuting x, y, z we know that $M_x = M_y = M_z$. It turns out that M_x is easiest to compute:

$$\begin{aligned} M_x &= \iiint_D x \, dx \, dy \, dz \\ &= \int_0^1 x \left(\int_0^{1-x} \left(\int_0^{1-x-y} 1 \, dz \right) dy \right) dx \\ &= \int_0^1 x \left(\int_0^{1-x} (1 - x - y) dy \right) dx \\ &= \int_0^1 x \left[y - xy - \frac{1}{2}y^2 \right]_0^{1-x} dx \\ &= \int_0^1 x \left[(1-x) - x(1-x) - \frac{1}{2}(1-x)^2 \right] dx \\ &= \int_0^1 x \left[\frac{1}{2}x^2 - x + \frac{1}{2} \right] dx \end{aligned}$$

$$\begin{aligned}
&= \int_0^1 \left[\frac{1}{2}x^3 - x^2 + \frac{1}{2}x \right] dx \\
&= \left[\frac{1}{2} \frac{1}{4}x^4 - \frac{1}{3}x^3 + \frac{1}{2} \frac{1}{2}x^2 \right]_0^1 \\
&= \frac{1}{2} \frac{1}{4} - \frac{1}{3} + \frac{1}{2} \frac{1}{2} \\
&= 1/24.
\end{aligned}$$

(c): The center of mass is

$$(\bar{x}, \bar{y}, \bar{z}) = \left(\frac{M_x}{M}, \frac{M_y}{M}, \frac{M_z}{M} \right) = \left(\frac{1/24}{1/6}, \frac{1/24}{1/6}, \frac{1/24}{1/6} \right) = \left(\frac{1}{4}, \frac{1}{4}, \frac{1}{4} \right).$$

Remark: Consider the solid n -dimensional “simplex” with $n + 1$ vertices:

$$(0, \dots, 0), (1, 0, \dots, 0), \dots, (0, \dots, 0, 1).$$

Using the same method, one can show that the n -dimensional “hypervolume” is $1/n!$ and the center of mass is $(1, 1, \dots, 1)/(n + 1)$. However, as you can imagine, the computation is messy.

Problem 5. Cylindrical Coordinates. Let D be a solid cone of radius 1 and height 1. We can think of this as the solid region defined by $x^2 + y^2 \leq 1$ and $0 \leq z \leq 1 - \sqrt{x^2 + y^2}$. Use cylindrical coordinates to compute the integral

$$\iiint_D z \, dx \, dy \, dz.$$

[Hint: Cylindrical coordinates are defined by $x = r \cos \theta$, $y = r \sin \theta$, $z = z$, and satisfy $\partial(x, y, z)/\partial(r, \theta, z) = r$. That is, $dx \, dy \, dz = r \, dr \, d\theta \, dz$.]

In cylindrical coordinates, the cone D has parametrization $0 \leq r \leq 1$, $0 \leq \theta \leq 2\pi$ and $0 \leq z \leq 1 - r$. Hence

$$\begin{aligned}
\iiint_D z \, dx \, dy \, dz &= \iiint_D z r \, dr \, d\theta \, dz \\
&= 2\pi \int_0^1 r \left(\int_0^{1-r} z \, dz \right) dr \\
&= 2\pi \int_0^1 r \left[\frac{1}{2} z^2 \right]_0^{1-r} dr \\
&= 2\pi \int_0^1 r(1-r)^2 dr \\
&= \pi \int_0^1 [r^3 - 2r^2 + r] dr \\
&= \pi \left[\frac{1}{4} r^4 - 2 \frac{1}{3} r^3 + \frac{1}{2} r^2 \right]_0^1 \\
&= \pi \left(\frac{1}{4} - \frac{2}{3} + \frac{1}{2} \right) \\
&= \pi/12.
\end{aligned}$$

Remark: If the cone has uniform density 1 then we just computed M_z . The volume of a cone is $(1/3)\pi(\text{radius})^2(\text{height}) = \pi/3$ and by symmetry we have $M_x = M_y = 0$, hence the center of mass of the cone is

$$(\bar{x}, \bar{y}, \bar{z}) = \left(\frac{M_x}{M}, \frac{M_y}{M}, \frac{M_z}{M} \right) = \left(0, 0, \frac{\pi/12}{\pi/3} \right) = \left(0, 0, \frac{1}{4} \right).$$

That is, the center of mass on the main axis at $1/4$ of the height. This same result holds for any cone of any radius and height.

Problem 6. Spherical coordinates ρ, ϕ, θ are defined by

$$x = \rho \sin \phi \cos \theta,$$

$$y = \rho \sin \phi \sin \theta,$$

$$z = \rho \cos \phi,$$

and satisfy $\partial(x, y, z)/\partial(\rho, \phi, \theta) = \rho^2 \sin \phi$. That is, $dx dy dz = \rho^2 \sin \phi d\rho d\phi d\theta$. Use spherical coordinates to compute the integral

$$\iiint_D \frac{1}{x^2 + y^2 + z^2} dx dy dz,$$

where D is the unit sphere. Even though the function $f(x, y, z) = 1/(x^2 + y^2 + z^2)$ goes to infinity when $(x, y, z) \rightarrow (0, 0, 0)$, the integral is still finite.

$$\begin{aligned} \iiint_D \frac{1}{x^2 + y^2 + z^2} dx dy dz &= \iiint_D \frac{1}{\rho^2} \rho^2 \sin \phi d\rho d\phi d\theta \\ &= \iiint_D \sin \phi d\rho d\phi d\theta \\ &= \int_0^{2\pi} 1 d\theta \int_0^\pi 1 d\rho \int_0^\pi \sin \phi d\phi \\ &= 2\pi [-\cos \phi]_0^\pi \\ &= 2\pi [-\cos(\pi) + \cos(0)] \\ &= 2\pi [-(-1) + (1)] \\ &= 4\pi. \end{aligned}$$

Remark: This looked like the hardest problem on HW4, but it was actually the easiest!

Remark: The analogous integral in one dimension is $\int_{-1}^1 (1/x^2) dx$, which diverges. The analogous integral in two dimensions also diverges:

$$\begin{aligned} \iint_{\text{unit disk}} \frac{1}{x^2 + y^2} dx dy &= \iint \frac{1}{r^2} r dr d\theta \\ &= \int_0^{2\pi} d\theta \int_0^1 \frac{1}{r} dr \\ &= 2\pi [\ln(1) - \ln(0)] \\ &= 2\pi [0 - (-\infty)] \\ &= \infty. \end{aligned}$$

For some reason the three dimensional version converges. We will observe the same type of phenomenon when we study gravity.